

Frank — AMC Drill M8 · route inspection

25 Jun 2026 | Targets your real gap (2018 Q24, M8). Walk every fence, return to start. Whole-number answers in km — show your working.

1. [5 marks] A rectangular field is 16 km by 12 km. Fences run along all four sides and along both diagonals, which cross at the centre (each diagonal is 20 km long). A farmer starts at one corner, walks along every fence at least once, and returns to that same corner. What is the shortest total distance, in kilometres, she can walk?

2. [5 marks] A rectangular field is 24 km by 10 km. Fences run along all four sides and along both diagonals, which cross at the centre (each diagonal is 26 km long). A farmer starts at one corner, walks along every fence at least once, and returns to that same corner. What is the shortest total distance, in kilometres, she can walk?

3. [5 marks] A rectangular field is 18 km by 24 km. Fences run along all four sides and along both diagonals, which cross at the centre (each diagonal is 30 km long). A farmer starts at one corner, walks along every fence at least once, and returns to that same corner. What is the shortest total distance, in kilometres, she can walk?

4. [5 marks] A rectangular field is 30 km by 16 km. Fences run along all four sides and along both diagonals, which cross at the centre (each diagonal is 34 km long). A farmer starts at one corner, walks along every fence at least once, and returns to that same corner. What is the shortest total distance, in kilometres, she can walk?

5. [5 marks] A rectangular field is 40 km by 30 km. Fences run along all four sides and along both diagonals, which cross at the centre (each diagonal is 50 km long). A farmer starts at one corner, walks along every fence at least once, and returns to that same corner. What is the shortest total

distance, in kilometres, she can walk?

Each is a code-verified twin of the real 2018 Q24. First step every time: count the fences at each corner — are any odd?